MACHINE PSYCHOLOGY RESEARCH PROGRAMME

Judged Wrong, Answered the Same: Judgment-Revision Decoupling in Iterative LLM Self-Review

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KEY RESULTS

0/63 answer changes across all review rounds

9 decoupling rounds: "partially wrong", answer restated

36/36 elicitation solutions correct on ten trap problems

60/63 review rounds stating a final answer, $133.33 every time

AT A GLANCE

- Three frontier models solved one arithmetic word problem, then reviewed their own output seven times in succession, each round stating a graded judgment and final answer.
- In 9 of the 63 rounds, a model judged its own solution "partially wrong", a grade defined as containing a meaningful error, and restated the identical answer in the same response.

INTRODUCTION

Iterative self-review is widely used on the assumption that a model which finds fault with its own work will go on to repair it. Self-Refine, Reflexion, chain-of-verification and constitutional critique pipelines expect the model to evaluate and, because it evaluates, to revise.

Prior evidence shows intrinsic self-correction does not reliably improve accuracy, but aggregate accuracy cannot say whether judgment was ever connected to the answer. This case series isolates that moment, requiring judgment and answer in the same response, every round.

METHODS

- **Stimulus.** One arithmetic word problem, a bat-and-ball variant with an added internal contradiction: exactly one consistent reading, one correct answer: $133.33.
- **Models and chains.** GPT-5.4, Claude 4.6 Sonnet and Gemini 3.1 Pro, three independent sessions each (nine chains), fresh conversations, temperature 0, no external tools.
- **Protocol.** Round 1: a structured six-step solution. Rounds 2 to 8: a fixed five-question review of its own preceding output, ending with a graded judgment, (a) completely correct to (d) completely wrong, and the final answer restated.

RESULTS IN DETAIL

- **Answers never moved.** All nine chains answered $133.33 in Round 1; an answer was stated in 60 of the 63 review rounds, and it was $133.33 every time.
- **The nine decoupling rounds.** Eight are Claude 4.6 Sonnet's and one is Gemini 3.1 Pro's. Each contains specific, argued charges against the solution, and none was followed by revision in any later round.
- **The refusal episode.** One GPT-5.4 chain refused the review task in Round 2, spent Rounds 3 and 4 judging its own refusal (d) completely wrong, and at Round 5 the original answer returned, unchanged.

Model	(a) Completely correct	(b) Mostly correct	(c) Partially wrong	(d) Completely wrong	Rounds
GPT-5.4	0	18	0	2*	21
Claude 4.6 Sonnet	0	13	8	0	21
Gemini 3.1 Pro	7	13	1	0	21
Total	**7 (11.3%)**	**44 (71.0%)**	**9 (14.5%)**	**2 (3.2%)**	**63**

Judgment distribution across the 63 review rounds (62 rounds contain a judgment; the exception is the refusal round). *Both (d) judgments were directed at a refusal text under review, not at a mathematical solution.

CONCLUSIONS

- Judgment and revision came apart: the evaluative act was performed in full, with reasons, and the number it condemned was repeated without change.
- Self-review output should be read as evaluative discourse whose connection to revision is an empirical question, not as evidence that revision has occurred.
- Evaluations of self-correcting systems should report judgment and revision as separate variables, rather than post-review accuracy alone, which cannot distinguish "reviewed and confirmed" from "condemned and repeated".

LIMITATIONS

- Single-problem case series: percentages describe these 63 rounds only and estimate no population rate; independent units are the nine chains, not 63 rounds; no inferential statistics are reported.
- No wrong-answer condition: the answer was correct, so the study cannot show the models would fail to revise one that needed revising.
- The protocol is demanding and repetitive, requiring fresh criticism every round; some negative judgments may be artefacts of that demand.
- Some provenance details (access dates, interface parameters) were not recorded.

WHY IT MATTERS

- Self-review is used as a trust signal: the model checked, so the answer is presumed more sound. These rounds show the signal in full, adverse judgment included, while the connection that would make it meaningful is absent.
- For agentic systems that loop self-evaluation internally, the refusal episode warns: a chain can spend a quarter of its rounds evaluating an interruption instead of the task, resuming with no trace in its final answer that anything went wrong.